

Predicting Polymers' Glass Transition Temperature by a Chemical Language Processing Model

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Abstract

We propose a chemical language processing model to predict polymers' glass transition temperature (T_g) through a polymer language (SMILES) embedding and recurrent neural network. This model only receives the SMILES strings of polymer repeat units as inputs. Results show that the trained model demonstrates reasonable prediction accuracy on unseen polymer T_g. The model is applied for high-throughput screening to identify high-temperature polymers.

2. Dataset

The T_g dataset used in this study is collected from the literature and the PolyInfo database. It contains 391 polymers with experimental T_g values ranging from −125 °C to 360 °C. Each polymer is represented by its repeat unit SMILES string. Data were split 80/20 into training and test sets.

The model achieves a root mean square error (RMSE) of 24.2 °C and a mean absolute error (MAE) of 17.8 °C on the test set. The coefficient of determination R² = 0.87.

Table 1. Representative polymers in the dataset with experimental T_g values.

#	Polymer name	Repeat unit SMILES	T _g (°C)	Source
1	Polystyrene (PS)	*CC*(c1ccccc1)*	100	PolyInfo
2	Poly(methyl methacrylate) (PMMA)	*CC(C)(C(=O)OC)*	105	PolyInfo
3	Poly(vinyl chloride) (PVC)	*CC(Cl)*	81	PolyInfo
4	Poly(ethylene terephthalate) (PET)	*OCC(=O)c1ccc(cc1)C(=O)O*	69	PolyInfo
5	Polyethylene (PE)	*CC*	−125	PolyInfo
6	Polypropylene (PP)	*CC(C)*	−10	PolyInfo
7	Poly(lactic acid) (PLA)	*OC(C)C(=O)*	55	Lit.
8	Polycarbonate (PC)	*OC(=O)Oc1ccc(C(C)(C)c2ccc(O)cc2)cc1*	147	PolyInfo
9	Poly(vinyl acetate) (PVAc)	*CC(OC(C)=O)*	28	PolyInfo
10	Nylon-6 (PA6)	*C(=O)CCCCCN*	47	Lit.
11	Polyimide (PI-PMDA-ODA)	*N1C(=O)c2ccc3c(c2)C(=O)N(*)*	360	PolyInfo
12	Poly(ether sulfone) (PES)	*Oc1ccc(cc1)S(=O)(=O)c1ccc(O)cc1*	220	PolyInfo
13	Polyacrylonitrile (PAN)	*CC(C#N)*	100	PolyInfo
14	Poly(vinyl alcohol) (PVA)	*CC(O)*	85	PolyInfo
15	Poly(butyl acrylate) (PBA)	*CC(C(=O)OCCCC)*	−54	PolyInfo

The training set contains 313 polymers (RMSE = 12.6 °C) and the test set 78 polymers. The model was trained for 100 epochs with a learning rate of 0.001 using Adam optimizer. Embedding dimension was set to 64 and the LSTM hidden size to 128.

High-throughput screening was performed on the PI1M database (1 million virtual polymers). Polymers with predicted T_g > 300 °C were identified as high-temperature candidates. A total of 12,340 polymers met this criterion, primarily aromatic polyimides, polysulfones, and polyether ketones.